BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Standards and Protocols

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What is SMTP used for?

2. Unsigned 8-bit values can reach 255. Does 248 + 16 overflow?
   A) Yes: the sum is 264
   B) No: the sum is 264
   C) No: the sum is 8
   D) Yes: the sum is 232

3. What is the Internet?

4. A browser asks for a school's web page. "A client must host a website before asking for it." Correct this. Which computer provides the page?

Today's objectives

Explain why agreed network standards are useful. Distinguish standards from communication protocols.

Key vocabulary - Read, cover, recall

Standard
Agreed technical specification allowing compatibility.

Protocol
Agreed rules for communication between devices.

Interoperability
Different hardware/software systems working together.

HTTP
Protocol for web communication.

BLACK PEN: Write one definition in your own words.

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Why standards matter

Standards are agreed rules/specifications that allow equipment and software from different manufacturers to communicate.

Protocols in action

A protocol provides rules for exchanging data. Different services need different protocol functions.

Different communication jobs

A web request uses a web protocol such as HTTP/HTTPS. TCP supports reliable ordered delivery, while IP handles addressing and routing.

Layering is an explanation

Communication tasks can be considered in stages/layers. OCR J277 does not require memorising names and functions of all TCP/IP model layers; focus on the required standards and protocol functions.

I Do - Worked model

A school purchases equipment from several manufacturers. Explain why standards and communication protocols help it work together. [4 marks]

- Different manufacturers need compatible rules if their devices are to communicate.
- Agreed standards [1] help equipment from different makers interoperate [1].
- Communication protocols define common rules for sending data [1], so requests can be interpreted correctly by devices [1].
- Four points: standard -> compatibility, protocol -> effective exchange.

We Do - Guided practice

1. A tablet from one company uses a website hosted by another. Explain why communication standards matter. [4 marks]

2. A client requests a web page. Explain the separate roles of HTTP and TCP. [4 marks]

3. Explain how TCP and IP work together when sending data between networks. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Write the second print instruction.

```
print("Request")
```

You Do - Independent exam questions

Q2. State one benefit of agreed networking standards. [1 mark]

Q3. A public-transport ticket kiosk communicates with a booking service made by a different manufacturer. Explain two benefits of both systems following agreed communication standards or protocols. [4 marks]

Q4. Explain how standards and HTTP, TCP and IP help successful network communication. [6 marks]

Exit Ticket - True or False?

OCR J277 requires memorising all four TCP/IP layer names and their functions.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Sending email from a client to a mail server and between mail servers.

2. A - Yes: the sum is 264

3. A worldwide collection of interconnected computer networks.

4. A web server stores website resources and sends them when clients request them. Access needs working servers and network services.

I Do - Explain the model

Different manufacturers need compatible rules if their devices are to communicate.

Agreed standards [1] help equipment from different makers interoperate [1].

Communication protocols define common rules for sending data [1], so requests can be interpreted correctly by devices [1].

Four points: standard -> compatibility, protocol -> effective exchange.

We Do - Model answers

1. Both devices follow agreed networking rules [1] even though built by different manufacturers [1], so they can exchange compatible data [1] and use network services [1].

2. HTTP sets rules for requesting/transferring web content [1] so client/server know which resource is needed [1]. TCP helps deliver data reliably [1] in correct order [1].

3. TCP supports reliable delivery [1] and correct ordering [1]; IP uses destination network addressing [1] and routing between networks [1].

Code First - Answer

print("Request")
print("Response")
Expected output: Request
Response

You Do - Self-mark

Q2: They allow equipment/software from different manufacturers to communicate/interoperate [1].

Q3: Both systems use the same agreed message rules [1], so requests/responses are understood despite different manufacturers [1]. Shared standards enable interoperability [1], avoiding a separate custom communication system for each manufacturer's equipment [1].

Q4: Standards support device compatibility [1] between different manufacturers [1]. HTTP defines web requests and responses [1]. TCP supports reliable arrival [1] and correct order [1]. IP supplies destination addressing/routing [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.